

A Difference-in-Differences Evaluation of the SEC Tick Size Pilot Program

Dr. Tyler Ransom

David Colindres

August 1, 2026

Abstract

Between October 2016 and September 2018, the SEC ran a rare randomized policy experiment in U.S. market structure: the Tick Size Pilot Program, which widened the minimum quoting (and for certain groups, trading) increment from \$0.01 to \$0.05 for roughly 1,200 small-capitalization stocks against a randomized control group, with assignments published as public lists. I evaluate the Pilot’s liquidity effects with measures computable entirely from public data: Amihud (2002) illiquidity, the Corwin-Schultz (2012) high-low spread estimator, dollar volume, and realized volatility. In a two-way fixed effects difference-in-differences design on 1,116 securities over 48 months, the wider tick increased Amihud illiquidity by 10.6% (log-point estimate 0.101, s.e. 0.029) and widened estimated spreads by 5.25 points (s.e. 1.18), with the largest price-impact effects in Test Group 1. An event study shows no differential pre-trends (joint $p = 0.72$) and a full reversal at the Pilot’s termination ($p = 0.49$). Causal-forest heterogeneity, though sloping in the predicted direction, is statistically indistinguishable from zero. The findings corroborate the SEC’s own assessment that the Pilot raised trading costs without delivering the liquidity gains its proponents predicted.

1 Introduction

1.1 Motivation & Research Question

The minimum price increment, or “tick size”, is a basic design parameter of an equity market: it sets the width of the price grid on which liquidity providers quote, thereby shaping market making economics. A wider tick guarantees a larger minimum spread, and so more per-share revenue for liquidity providers, also raising the cost of immediacy paid by demanders. Whether widening the tick for small, thinly traded stocks would improve liquidity by subsidizing market making or harm it by taxing every trade was the question behind the SEC’s Tick Size Pilot Program, which grew out of the 2012 JOBS Act debate

over declining small-company market quality. Unlike most market-structure changes, which hit all securities at once and leave no control group, the Pilot randomly assigned roughly 2,400 eligible small caps to a control group and three treatment groups, and published the assignment lists as public files ([FINRA, 2016](#)). That makes it a true policy experiment, rare in finance, and a natural setting for difference-in-differences (DiD) estimation.

We ask the following question: Did widening the minimum tick to five cents improve or harm liquidity for treated small-cap stocks, and for which stocks was the effect largest? I answer this question using only public daily data, showing that the Pilot’s headline effects, documented in the literature with proprietary intraday data, are detectable with low-frequency, reproducible proxies. In 2024 the SEC adopted amendments moving in the opposite direction, halving the tick to \$0.005 for tick-constrained stocks, so causal evidence on how quoting increments shape small-cap liquidity bears directly on current rulemaking.

1.2 Preview of Findings & Method

I build a monthly panel of liquidity outcomes from October 2015 to September 2019, spanning a pre-period two-year treatment window, and a post-termination period. The identification strategy is a two-way fixed effects DiD: stock fixed effects absorb time-invariant differences, calendar-month fixed effects absorb market-wide shocks, and the coefficient of interest is treatment status interacted with the Pilot window. Randomized assignment makes parallel trends unusually credible, and I verify it with an event study. The Pilot also concluded on a known date, so I additionally test whether effects reverse at termination, a falsification exercise rarely available in policy evaluation. Finally, I estimate heterogeneous treatment effects with a causal forest ([Wager and Athey, 2018](#)), asking whether the tick change disproportionately harmed low-priced, tick-constrained stocks.

Treated stocks became more expensive to trade: Amihud price impact rose by 10.6% (log-point estimate 0.101, s.e. 0.029), estimated spreads widened by 5.25 points, dollar volume fell 6.1%, and realized volatility rose by 0.016. Both identification checks pass, with no differential pre-trends (joint $p = 0.72$) and effects that vanish once the wide tick is removed (joint $p = 0.49$). Heterogeneity across stock types faces in the predicted direction but is statistically undetectable.

Overall, Section 2 reviews the literature; Section 3 the Pilot’s design; Section 4 the data; Section 5 the empirical strategy; Section 6 the results and robustness; Sections 7-8 discuss policy and conclude. The appendix pipeline reproduces every number, table, and figure.

2 Literature Review

Direct evaluations of the Pilot. [Chung et al. \(2020\)](#), using consolidated limit order books, find the Pilot raised quoted and effective spreads by 15 basis points on average, increasing small-order execution costs, while liquidity for large orders improved. [Griffith and Roseman \(2019\)](#) reach a related but non-identical conclusion: displayed depth rose but overall trading costs increased, and the two papers disagree on liquidity. Even with rich data, the sign of the liquidity effect depends on the dimension measured. [Albuquerque et al. \(2020\)](#) show the Pilot’s liquidity shock had measurable price effects, consistent with liquidity being priced. The SEC’s staff assessment ([SEC DERA, 2018](#)) concludes the wider tick worsened market quality on average, which contributed to the Commission letting the Pilot lapse. [Rindi and Werner \(2019\)](#) use the three-group structure to separate mechanisms, documenting that the G3 trade-at provision shifted volume from off-exchange venues onto lit exchanges; tick policy and venue competition interact, a theme [Comerton-Forde et al. \(2019\)](#) develop by showing a tick’s value depends on exchange fee structures. Overall, the evaluations agree that small-order execution costs rose, but disagree about depth and efficiency, and report almost exclusively average effects.

Low-frequency liquidity measurement. [Amihud \(2002\)](#) proposes the ratio of absolute return to dollar volume as a price-impact measure, still the workhorse illiquidity proxy in asset pricing. [Corwin and Schultz \(2012\)](#) derive a bid-ask spread estimator from daily high and low prices and show it correlates highly with TAQ-based effective spreads, the object the intraday Pilot literature studies. Using both lets this paper test whether the Pilot’s documented intraday effects are recoverable from free data.

Methodology. [Bertrand et al. \(2004\)](#) show serially correlated outcomes require clustering standard errors at the stock level. Given the test groups phased in over a few weeks in October 2016, the design is effectively single-date treatment, avoiding the staggered-adoption pitfalls of [Callaway and Sant’Anna \(2021\)](#), whose estimator I apply as robustness. For heterogeneity, [Wager and Athey \(2018\)](#) develop causal forests with valid CATE inference; the forest’s average effects use the doubly robust AIPW construction of [Chernozhukov et al. \(2018\)](#).

This paper, instead of overturn intraday evidence, does the following: replicates the Pilot’s headline liquidity findings with free daily data alone, exploits the termination as a built-in reversal test, and characterizes heterogeneity with causal forests, which the early Pilot literature largely didn’t do.

3 The Tick Size Pilot

The Pilot ran from October 3, 2016 through September 28, 2018 under a National Market System plan ([SEC DERA, 2018](#); [FINRA, 2016](#)). Eligible securities were common stocks

with market capitalization of \$3 billion or less, price of at least \$2.00, and average daily volume of one million shares or less. Approximately 2,400 securities were stratified and randomly assigned:

- **Control** ($\approx 1,200$): continued to quote and trade in \$0.01 increments.
- **Test Group 1** (≈ 400): quoted in \$0.05 increments; traded at any increment.
- **Test Group 2** (≈ 400): quoted and traded in \$0.05 increments, with midpoint, retail price improvement, and negotiated-trade exemptions.
- **Test Group 3** (≈ 400): G2 requirements plus a “trade-at” rule restricting dark-pool and internalized executions.

The structure separates coarser quoting (G1), coarser trading (G2), and off-exchange restrictions (G3). Firms could not opt in or out, and no securities were added mid-Pilot, limiting selection concerns. At termination on October 1, 2018 all test-group securities reverted to penny ticks, creating a second, equally sharp event.

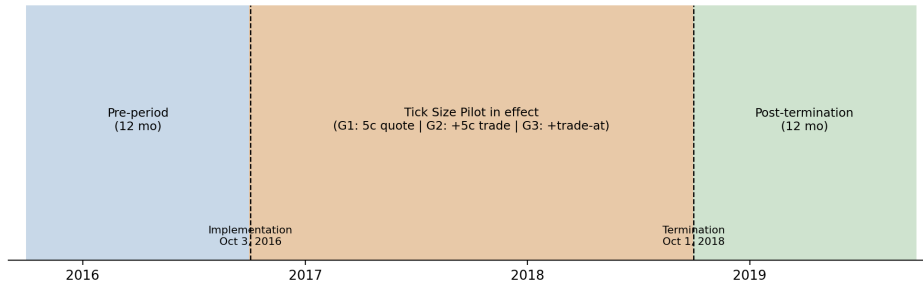


Figure 1: Timeline of the Tick Size Pilot and estimation windows.

4 Data

4.1 Treatment Assignment

Treatment status is reconstructed from FINRA’s cumulative Pilot Changes file (`TSPilotChanges.txt`), which logs every assignment event (ticker, name, group, effective date, deletion date) and which FINRA still serves publicly. The companion daily securities file was cleared post-Pilot, so the changes file is the official FINRA source that survives (FINRA, 2016). Given the test groups phased in over five Monday waves in October 2016, each security’s assignment is the group of its record active on November 1, 2016. Exchange test symbols (ex., NTEST, ZAZZT) are excluded. The baseline sample drops securities that switched groups or tickers after the reference date. The assignment sample contains 1,186 control, 398 G1, 397 G2, and 395 G3 securities, with 134 flagged switchers.

4.2 Market Data

Daily OHLC, adjusted close, and volume come from Yahoo Finance (`yfinance`), October 1, 2015 to October 1, 2019. Delisted tickers may be missing or truncated, and Yahoo restates history for all splits through the present, so securities with large subsequent reverse splits display early-sample closes far above traded prices. All liquidity outcomes are invariant to this restatement; dollar volume as volume adjusts inversely to price, Corwin-Schultz since split factors cancel in high-low ratios, and returns given they use adjusted closes. Only the price level is on the adjusted basis. Of the assignment sample, 1,137 securities (47.9%) have usable histories; coverage is indistinguishable across groups (49.0%, 46.2%, 44.8%, 49.1% for C, G1, G2, G3; $\chi^2 p = 0.435$).

4.3 Liquidity Measures

Outcomes are computed at the stock-month level. **Amihud illiquidity**: for stock i in month t with D_{it} valid days,

$$\text{ILLIQ}_{it} = \frac{1}{D_{it}} \sum_{d=1}^{D_{it}} \frac{|r_{i,d}|}{P_{i,d} V_{i,d}} \times 10^6, \quad (1)$$

where $r_{i,d}$ is the daily return and $P_{i,d} V_{i,d}$ dollar volume (Amihud, 2002); higher values mean a dollar of volume moves price more, and regressions use $\log(\text{ILLIQ}_{it})$. **Corwin-Schultz spread**: following Corwin and Schultz (2012), for each two-day window

$$\beta = \sum_{j=0}^1 \left[\ln \left(\frac{H_{d+j}}{L_{d+j}} \right) \right]^2, \quad \gamma = \left[\ln \left(\frac{H_{d,d+1}}{L_{d,d+1}} \right) \right]^2, \quad (2)$$

$$S_d = \frac{2(e^\alpha - 1)}{1 + e^\alpha}, \quad \alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma}{3 - 2\sqrt{2}}}, \quad (3)$$

with negative two-day estimates set to zero before monthly averaging; S_{it} is in basis points. I also compute log dollar volume, realized volatility, and the zero-return-day share.

4.4 Sample Construction & Descriptives

The window runs October 2015 through September 2019. Cleaning proceeds in six steps: (1) observations with non-positive prices or high below low are dropped as feed errors; (2) closes above \$100,000 are excluded as retroactive split-adjustment artifacts; (3) Corwin-Schultz estimates are discarded for two-day windows spanning a raw close-to-close move above 20%, where the estimator's variance decomposition is invalid; (4) stock-months require at least 15 valid days; (5) outcomes and price are winsorized at the 1st/99th percentiles within month; (6) baseline drops flagged switchers. Returns use adjusted closes;

dollar volume uses unadjusted closes, reflecting [Amihud \(2002\)](#).

Table 1 reports pre-period balance. Consistent with randomization, the groups are indistinguishable on log dollar volume, log illiquidity, volatility, and zero-return share; adjusted-basis price ($p = 0.048$) and the CS spread ($p = 0.058$) show marginal differences. Two near-threshold rejections among six tests are unremarkable, and level differences are absorbed by the stock fixed effects in any case. Figure 2 plots mean log illiquidity by group; the four series track each other closely through large common swings, consistent with parallel trends. Level differences persist but are absorbed by stock fixed effects, and the treated-control gap during the Pilot is modest relative to the common variation, so identification comes from the fixed-effects comparison than from raw means.

Table 1: Pre-period summary statistics and balance by pilot group

	Control	Test Group 1	Test Group 2	Test Group 3	p (equal)
Price (\$)	46.82	29.67	37.32	27.02	0.048
log \$Vol	13.99	13.96	14.23	14.05	0.585
log ILLIQ	-4.04	-4.03	-4.35	-4.10	0.513
CS spread (bp)	94.07	88.13	85.63	86.54	0.058
RVol	0.41	0.38	0.39	0.39	0.206
Zero-ret share	0.05	0.06	0.05	0.05	0.825
Securities	571	177	170	190	

Means of stock-level averages over the twelve pre-Pilot months; final column reports the F -test p -value for equality across groups. Sample: 1,116 securities (1,108 with pre-Pilot observations). Data: FINRA pilot lists; Yahoo Finance daily data.

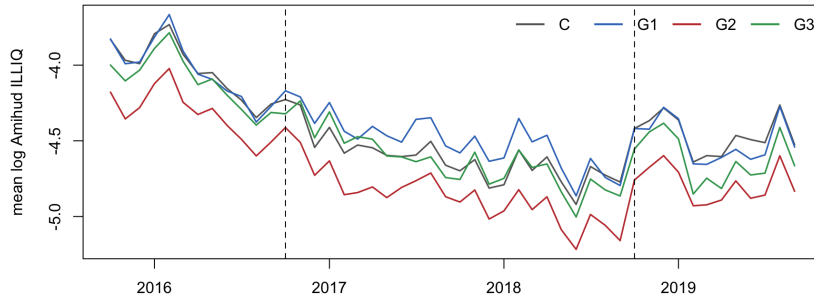


Figure 2: Mean log Amihud illiquidity by pilot group, monthly. Vertical lines mark implementation (October 2016) and termination (October 2018).

5 Empirical Strategy

5.1 Baseline Difference-in-Differences

The baseline is a two-way fixed effects DiD on the stock-month panel:

$$y_{it} = \alpha_i + \lambda_t + \beta (\text{Treat}_i \times \text{Pilot}_t) + \varepsilon_{it}, \quad (4)$$

with stock fixed effects α_i , month fixed effects λ_t , $\text{Treat}_i = 1$ for any test group, and $\text{Pilot}_t = 1$ during October 2016-September 2018; under parallel trends β is the average treatment effect on the treated. To separate mechanisms,

$$y_{it} = \alpha_i + \lambda_t + \sum_{g \in \{1,2,3\}} \beta_g (\mathbf{1}[G_i = g] \times \text{Pilot}_t) + \varepsilon_{it}, \quad (5)$$

where β_1 isolates the quoting constraint, $\beta_2 - \beta_1$ the trading increment, and $\beta_3 - \beta_2$ the trade-at rule. Standard errors are clustered by stock throughout (Bertrand et al., 2004).

5.2 Event Study & the Termination Test

Dynamics come from

$$y_{it} = \alpha_i + \lambda_t + \sum_{\substack{k=-12 \\ k \neq -1}}^K \delta_k (\text{Treat}_i \times \mathbf{1}[t - t_0 = k]) + \varepsilon_{it}, \quad (6)$$

with event time k in months from October 2016 and $k = -1$ omitted. Parallel trends implies $\delta_k \approx 0$ for $k < 0$; the post-September-2018 coefficients provide a reversal test: a causal effect should attenuate toward zero once the tick reverts, while identical persistence suggests confounding.

5.3 Identification

Causality rests on randomized assignment, no opt-in or opt-out, and identical eligibility screens for all groups. The main threats include spillovers (trading migrating to control stocks would bias volume effects, and Chung and Chuwonganant (2023) document such migration directly), attrition (delisted stocks exit Yahoo deterministically), and measurement.

5.4 Heterogeneous Effects: Causal Forest

Theory predicts the constraint binds most for low-priced, previously tight-spread stocks. Rather than pre-committing to interaction terms, I estimate CATEs with a causal forest

(Wager and Athey, 2018) on the collapsed two-period dataset, using pre-period covariates: price, log dollar volume, illiquidity, CS spread, volatility, and zero-return share. Average effects use the forest’s doubly robust AIPW construction (Chernozhukov et al., 2018). The question is whether $\tau(x)$ declines in the pre-period spread, whether tick-constrained stocks bear the largest cost.

6 Results

6.1 Baseline Estimates

Table 2 reports Equation 4. The Pilot raised log Amihud illiquidity by 0.101 (s.e. = 0.029), roughly 10.6%, and widened the CS spread by 5.25 points (s.e. 1.18); dollar volume fell 6.1% and realized volatility rose 0.016 (s.e. = 0.004). Relative to a pre-period treated mean spread of 87 points, the spread effect is a 6% increase in round-trip trading costs, and each dollar traded moved prices 11% more. For market makers, the guaranteed wider spread raised revenue per share but didn’t attract offsetting liquidity provision, as volume fell rather than rose.

Table 2: Baseline difference-in-differences estimates

	log ILLIQ	CS spread (bp)	log \$Vol	RVol
Treat $\times$ Pilot	0.101*** (0.029)	5.248*** (1.180)	-0.063** (0.026)	0.016*** (0.004)
Stock FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Obs.	53,264	53,287	53,287	53,287

Each column estimates Equation 4 with stock and month fixed effects; standard errors clustered by stock in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: 1,116 stocks; 53,264-53,287 stock-months across outcomes.

6.2 Effects by Test Group

Table 3 reports Equation 5. For price impact the ordering is G1 (0.146***) > G3 (0.100*) > G2 (0.056, n.s.), while for the spread proxy G1 (5.99***) and G2 (5.89***) are nearly identical and G3 (3.98*) is smallest. First, the quoting constraint alone is sufficient to generate effects: adding the trading increment in G2 changes the spread estimate by essentially nothing. Second, the price-impact ordering does not reproduce the intraday literature’s finding that the trade-at group fared worst; if anything G3’s trade-at rule, by forcing volume onto displayed venues, appears to have cushioned price impact, and G2’s midpoint and retail exemptions may explain its muted response. Daily Amihud captures price impact, whereas the intraday studies measure quoted and effective spreads.

Table 3: Effects by test group

	log ILLIQ	CS spread (bp)	log \$Vol	RVol
G1 $\times$ Pilot	0.145*** (0.040)	5.990*** (1.649)	-0.086** (0.035)	0.021*** (0.006)
G2 $\times$ Pilot	0.056 (0.040)	5.887*** (1.658)	-0.018 (0.037)	0.015*** (0.006)
G3 $\times$ Pilot	0.100** (0.040)	3.980** (1.683)	-0.082** (0.038)	0.010* (0.006)
Stock FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Obs.	53,264	53,287	53,287	53,287

Estimates of Equation 5. Stock and month fixed effects included; standard errors clustered by stock.

6.3 Event Study & Termination

Figure 3 plots the δ_k for log illiquidity. Pre-period coefficients are indistinguishable from zero (Wald $p = 0.723$) and post-termination coefficients are jointly zero (Wald $p = 0.491$). Coefficients hover around zero for all twelve pre-months, turn positive within the first months of the Pilot, build to 0.12 log points mid-window, then settle to around zero post-termination. Since control stocks experienced every market-wide shock in this window, the only event coinciding with both the switch-on and the switch-off is the tick change.

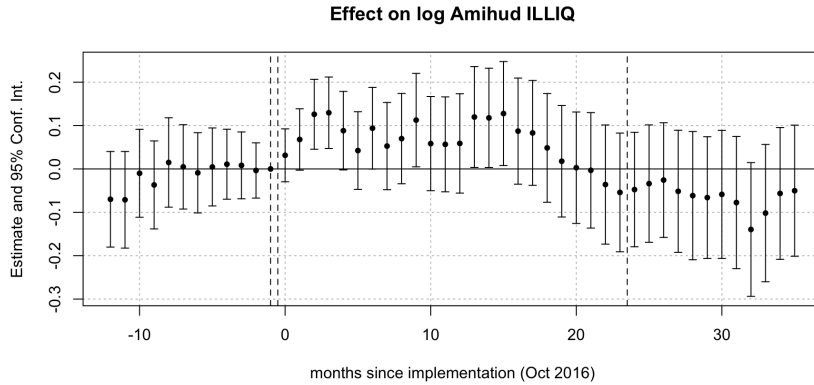


Figure 3: Event-study estimates of Equation 6 for log Amihud illiquidity with 95% confidence intervals (stock-clustered); omitted period is the month before implementation. Vertical lines mark implementation and termination.

6.4 Heterogeneous Effects

Figure 4 summarizes the causal forest. The average effect from the collapsed design is 0.064 (s.e. 0.046), positive but not significant, and no coefficient on pre-period characteristics is distinguishable from zero. The point pattern does slope as tick-constraint economics predict, with effects largest for tight pre-period spreads and fading, even negative, for

stocks already quoting far above five cents; but with roughly 1,100 stocks and daily proxies the projection cannot reject a uniform effect. I read this as a power limitation, not evidence against the theory.

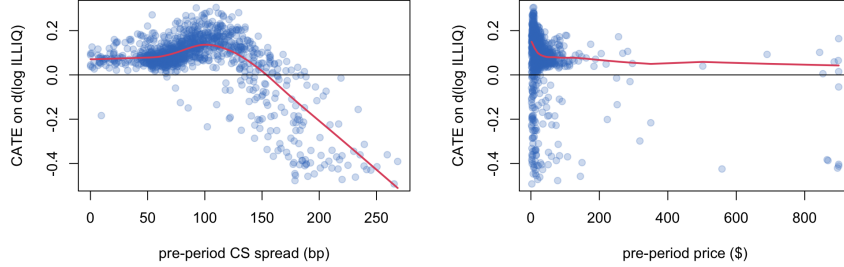


Figure 4: Conditional average treatment effects on log Amihud illiquidity from a causal forest, against pre-period Corwin–Schultz spread (left) and share price (right).

6.5 Robustness

Table 4 reports four checks: intention-to-treat retaining switchers, a balanced panel addressing attrition, dropping G3, whose trade-at rule most plausibly generated spillovers, and the Callaway and Sant’Anna (2021) estimator on the Pilot window only, since it assumes absorbing treatment. 0.106 with switchers retained, 0.099 in the balanced panel, and 0.102 with G3 excluded, are all significant and within a standard error of the baseline. The Callaway–Sant’Anna estimate is smaller at 0.059 with a wider, bootstrapped standard error. The baseline point estimate lies within its 95% confidence interval, so the two estimators are statistically consistent.

Table 4: Robustness of the baseline estimate (log Amihud illiquidity)

Specification	Treat $\times$ Pilot	(s.e.)	Obs.
ITT (keep switchers)	0.106	(0.029)	54,058
Balanced panel	0.099	(0.029)	52,729
Drop G3	0.102	(0.032)	44,151
Callaway–Sant’Anna (2021)	0.059	(0.043)	39,915

Each row re-estimates Equation 4 under the indicated specification. Stock and month fixed effects; standard errors clustered by stock (Callaway–Sant’Anna row uses the multiplier bootstrap).

7 Policy Discussion

The Pilot was motivated by the hypothesis that wider ticks would subsidize small-cap market making and ultimately support IPOs. The SEC’s assessment concluded the opposite on average, and the Commission let the Pilot lapse (SEC DERA, 2018). The estimates

here, from daily public data alone, corroborate that conclusion: on every margin visible in daily data, the wider tick made treated small caps worse to trade, with about 11% more price impact per dollar traded, 5 points more spread, and 6% less volume. Theory predicts the cost falls hardest on previously tick-constrained stocks, and while my heterogeneity estimates likewise, they cannot separate the burden across stock types with daily data. That caution still supports the direction the SEC ultimately took in 2022-2024 rulemaking toward finer, spread-calibrated ticks. Given assignment was public and randomized, the Pilot’s failure could be measured credibly and cheaply, which argues for building randomized rollouts into future market-structure changes.

Several limitations still arise. Daily proxies cannot separate displayed from hidden liquidity, small-order from large-order costs, or maker from taker economics, margins on which the intraday evidence is mixed [Chung et al., 2020](#). Yahoo data are survivorship-prone, and differential delisting would bias estimates. Spillovers toward control stocks would bias the volume effect toward overstatement, since migrating activity raises the control counterfactual. The estimates are therefore best read as the Pilot’s effect on daily-data-visible liquidity for surviving securities.

8 Conclusion

The Tick Size Pilot offered a rare randomized experiment on a first-order market design parameter. Using a DiD design on monthly liquidity measures built entirely from public daily data, I find that widening the tick to five cents raised the price impact of trading by roughly 11% and estimated spreads by 5 points while volume fell 6%, with the quoting constraint alone sufficient to generate the spread effect and no statistically detectable concentration of the burden across observable stock types. The results align with the proprietary-data literature and the SEC’s own assessment, demonstrating credible market-structure evaluation is feasible with free data when treatment assignment is randomized/public. Future work could prompt designing price-efficiency outcomes, formally model delisting attrition, or apply the same toolkit to the SEC’s 2024 move to half-cent ticks.

References

- Albuquerque, R., Song, S., and Yao, C. (2020). The Price Effects of Liquidity Shocks: A Study of the SEC’s Tick Size Experiment. *Journal of Financial Economics*, 138(3):700–724.
- Amihud, Y. (2002). Illiquidity and Stock Returns: Cross-Section and Time-Series Effects. *Journal of Financial Markets*, 5(1):31–56.
- Bertrand, M., Duflo, E., and Mullainathan, S. (2004). How Much Should We Trust Differences-in-Differences Estimates? *Quarterly Journal of Economics*, 119(1):249–275.
- Callaway, B. and Sant’Anna, P.H.C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2):200–230.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/Debiased Machine Learning for Treatment and Structural Parameters. *Econometrics Journal*, 21(1):C1–C68.

- Chung, K.H. and Chuwonganant, C. (2023). Tick Size and Price Efficiency: Further Evidence from the Tick Size Pilot Program. *Financial Management*, 52(3):483–511.
- Chung, K.H., Lee, A.J., and Rösch, D. (2020). Tick Size, Liquidity for Small and Large Orders, and Price Informativeness: Evidence from the Tick Size Pilot Program. *Journal of Financial Economics*, 136(3):879–899.
- Comerton-Forde, C., Grégoire, V., and Zhong, Z. (2019). Inverted Fee Structures, Tick Size, and Market Quality. *Journal of Financial Economics*, 134(1):141–164.
- Corwin, S.A. and Schultz, P. (2012). A Simple Way to Estimate Bid-Ask Spreads from Daily High and Low Prices. *Journal of Finance*, 67(2):719–760.
- Financial Industry Regulatory Authority (2016). Tick Size Pilot Program: Data Collection Securities and Pilot Securities Files. <https://www.finra.org/rules-guidance/key-topics/tick-size-pilot-program>.
- Griffith, T.G. and Roseman, B.S. (2019). Making Cents of Tick Sizes: The Effect of the 2016 U.S. SEC Tick Size Pilot on Limit Order Book Liquidity. *Journal of Banking & Finance*, 101:104–121.
- Rindi, B. and Werner, I.M. (2019). U.S. Tick Size Pilot. Fisher College of Business Working Paper No. 2017-03-018, The Ohio State University (SSRN 3041644).
- U.S. Securities and Exchange Commission, Division of Economic and Risk Analysis (2018). Tick Size Pilot Plan and Market Quality. DERA White Paper. https://www.sec.gov/files/dera_wp_tick_size-market_quality.pdf.
- Wager, S. and Athey, S. (2018). Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests. *Journal of the American Statistical Association*, 113(523):1228–1242.

Appendix A: Code & Reproducibility

All results are produced by the R pipeline via a single command, `Rscript R/00_run_all.R`, executing six scripts: 01 reconstructs treatment assignment from the FINRA changes file; 02 downloads daily Yahoo data; 03 builds the monthly liquidity panel (Equations 1–3); 04 estimates the balance table, baseline, group-wise, event-study, and robustness specifications (Tables 1–4, Figures 2–3); 05 fits the causal forest (Figure 4); 06 writes every headline estimate to a transcription sheet. A knittable notebook (`tsp_pipeline.Rmd`) runs the identical analysis; a parallel Python pipeline in `src/` reproduces Tables 1–3 and Figures 2–4 via `econml`’s `CausalForestDML`. Every table and figure is written directly to `paper/` by the code block that produces it.

Appendix B: Variable Definitions

ILLIQ: Amihud (2002) illiquidity, Equation 1, monthly mean of daily $|r|/\text{dollar volume} \times 10^6$, logged in regressions. **CS spread**: Corwin–Schultz (2012) two-day estimator, Equations 2–3, negatives set to zero, monthly mean, in basis points. **log \$Vol**: log of monthly mean daily dollar volume (`close` $\times$ shares). **RVol**: annualized standard deviation of daily returns within month. **ZeroRet**: share of trading days with a zero return. **Treat**: assignment to G1, G2, or G3 in the FINRA pilot file. **Pilot**: calendar months October 2016 through September 2018.